

Mapping and breeding value evaluation of a semi-dominant semi-dwarf gene in upland rice

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ABSTRACT

Plant height is an important trait related to yield potential and plant architecture. A suitable plant height plays a crucial role in improvement of rice yield and lodging resistance. In this study, we found that the traditional upland landrace 'Kaowenghan' (KWH) showed a special semi-dwarf phenotype. To identify the semi-dwarf gene from KWH, we raised BC₂F₄ semi-dwarf introgression lines (IL) by hybridization of the japonica rice cultivar 'Dianjingyou1' (DJY1) and KWH in a DJY1 background. The plant height of the homozygous semi-dwarf IL (IL-87) was significantly reduced compared with that of DJY1. The phenotype of the F₁ progeny of the semi-dwarf IL-87 and DJY1 showed that the semi-dwarf phenotype was semi-dominant. QTL mapping indicated that the semi-dwarf phenotype was controlled by a major QTL *qDH1* and was localized between the markers RM6696 and RM12047 on chromosome 1. We also developed near-isogenic lines (NIL) from the BC₃F₃ population, and found that the yield of homozygous NIL (NIL-2) was not significantly different compared to DJY1. Breeding value evaluation through investigation of the plant height of the progeny of NIL (NIL-2) and cultivars from different genetic background indicate that the novel semi-dwarf gene shows potential as a genetic resource for rice breeding.

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Rice (*Oryza sativa*) is one of the most important cereal crops and is a staple food in many countries worldwide. With the rapid increase in the global population and the emergence of a food crisis, increasing rice yields has become the focus of rice research and breeding programs (Khush, 2005). Plant height is an important rice trait that directly affects the yield and lodging resistance. Dwarf plants show enhanced lodging resistance, but an excessively dwarfed plant leads to inadequate growth and a tendency for densely overlapping leaves, thus reducing rice yield. However, extremely tall plants are susceptible to lodging which also reduces crop yield and quality (Zhang et al., 2017b). The 'Green Revolution' and the breeding of hybrid rice cultivars have significantly improved rice yield by utilization of semi-dwarf gene resources

(Hedden, 2003; Khush, 1999). Therefore, the discovery and utilization of semi-dwarf genes are of considerable importance for the improvement of rice yield.

In recent years, many genes that modulate plant height, including more than 60 recessive dwarf genes and 10 recessive semi-dwarf genes, have been identified or cloned (Zhang et al., 2014). *DWARF 18* (*D18*) is a dwarf gene that encodes OsGA3ox2, which controls rice plant height by participating in gibberellins (GA) biosynthesis (Tong et al., 2014; Iwamoto et al., 2011; Itoh et al., 2001). The dwarf gene *gibberellin insensitive dwarf 1* (*gid1*) encodes a soluble gibberellin receptor that affects plant height by mediating GA signaling transduction in rice (Ayano et al., 2014; Tanaka et al., 2006; Ueguchi-Tanaka et al., 2005). *Brassinosteroid-deficient dwarf 1* (*brd1*) encodes a key enzyme involved in rice brassinolide (BR) biosynthesis, and the mutant for which is characterized by a severely dwarfed stem, curved and deformed leaves, crown root, small ear shape and grain size (Mori et al., 2002; Hong et al., 2002). The gene *D61*, which encodes a rice BR receptor kinase involved in BR signaling transduction, reduces plant height (Zhang et al., 2016; Nakamura et al., 2006; Yamamuro et al., 2000). *DWARF*

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KAMIKAWABUNWAI TILLERING (HTD2, also known as D88 or D14), a component of the strigolactones (SLs) signaling pathway, encodes an esterase that is involved in inhibition of branching and negatively regulates the number of tillers (Kagiyama et al., 2013; Liu et al., 2009). WEALTHY FARMER'S PANICLE (WFP, also known as IPA1) encodes a transcription factor that can directly bind to the promoter of the negative regulator FINE CULM 1 (FC1), which controls the growth of tiller lateral buds and thus inhibits tillering. WFP also affects plant height and panicle length by directly regulating the important gene DENSE AND ERECT PANICLE 1 (DEP1), thereby determining the spike type (Zhang et al., 2017a; Lu et al., 2013; Miura et al., 2010).

In contrast to recessive genes, few dominant dwarf genes have been reported. In rice, only a few dominant or semi-dominant dwarf genes, including D53, Ssi1, Sdd(t), Dx, TID1, LB4D, Slr-d, D-h, SBI, Dd7, Sdt97, have been reported (Liu et al., 2008, 2018; Zhao et al., 2018b; Piao et al., 2014; Liang et al., 2011; Asano et al., 2009; Miura et al., 2009; Qin et al., 2008; Tong et al., 2007; Wei et al., 2006; Sunohara et al., 2003). However, dominant dwarf genes have special advantages in crop breeding (Qin et al., 2008). For instance, unless both parents carry the same recessive semi-dwarf gene, recessive genes cannot be used in hybrid rice cultivars. Dominant dwarf alleles, in contrast, can be used when only one parent carries the dominant allele. Thus, the germplasm of the other parent can be widely chosen, sharply reducing time, cost, and labor in breeding work (Qin et al., 2008; Liang et al., 2004). Therefore, it is important to discover and utilize novel dwarf genes, especially dominant genes for breeding dwarf varieties.

The dwarf rice germplasms widely used in commercial production in China include 'Aijiaonante', 'Aizaizhan', 'Dee-geo-woogen', which genetic analysis has shown all carry the recessive gene semi-dwarf 1 (*sd1*) (Chen et al., 2013). The frequent use of limited dwarfism sources may disadvantage the diversification of rice varieties, hinder the genetic improvement process, and cause a 'bottleneck' effect in the genetic background available for developing new rice varieties (Matsuo et al., 1997; Luh, 1980; Hargrove and Cabanilla, 1979). Thus, more useful dwarfism sources are required for rice breeding.

In the present study, we found a unique upland landrace 'Kao-wenghan' (KWH) from Yunnan province, which unlike other upland rice landraces shows a semi-dwarf phenotype. Genetic analysis indicated that the semi-dwarf phenotype is semi-dominant. To identify a potentially novel dominant dwarf gene for rice breeding, we crossed KWH donor male parents with DJY1 to generate homozygous semi-dwarf IL-87 and a mapping population. QTL mapping and breeding value evaluation indicate that we identified a novel semi-dwarf gene which may play an important role in the study of the genetic mechanisms of dominant dwarf and breeding utilization in rice.

1. Materials and methods

1.1. Plant materials

DJY1 is a *japonica* rice cultivar, and KWH is a unique semi-dwarf landrace of *japonica* upland rice; both are grown in Yunnan province, China. Progeny of the F₁ were obtained by crossing DJY1 with KWH; the F₁ individuals were consecutively backcrossed to DJY1 as male and recurrent parent until BC₂F₁; semi-dwarf IL-87 were developed from the BC₂F₄ population. The BC₃F₂ population (2015H3E483), which comprised 68 plants, were developed for genetic analysis and preliminary QTL mapping by crossing DJY1 (male parent) and the semi-dwarf IL-87 (female parent). The BC₃F₃ population (2017J1E87), including 396 individuals developed from the heterozygous BC₃F₂ plants, were planted for further QTL

mapping. To confirm the effect of QTL, we also developed QTL homozygous NIL(NIL-2) from the BC₃F₃ populations for phenotypic assessment.

For breeding value evaluation, F₁ progeny were produced by crossing the NIL-2 with 20 core rice cultivars obtained from the International Rice Research Institute (IRRI). These cultivars are genetically diverse and actively used in international breeding programs on account of their wide range of agronomic attributes, and belong to three genetic background groups: *Japonica*, *Aus* and *Indica* (McNally et al., 2009) (Table S1). The plant height of F₁ plants was investigated.

All materials were grown in a paddy field at an experimental station at the Xishuangbanna Tropical Botanical Garden, located in Jinghong, Yunnan Province, China. All F₁ and parents were sequentially planted in three rows for each material. DJY1, KWH, IL-87 and NIL-2 were planted in a randomized complete block design with three repeats, each block contained 4 rows with 10 plants per row. All the material was planted within a 27 cm × 15 cm plant space. Agronomic traits were examined for eight randomly selected plants from the center of the block. Field management, including irrigation, fertilization, and pest control followed standard agricultural practices.

1.2. Phenotypic measurements

Plant height (cm) was measured from the ground to the tip of the tallest panicle (awns excluded) at full maturity (Würschum et al., 2015). Panicle length (cm) was measured from the base of the panicle to the tip of panicle (Zhao et al., 2018a). The main tiller of each plant was selected to measure the internode length (cm); n1 was measured from the base of the panicle to the first node below; n2 was measured from the first to the second node, and so on (Wang et al., 2017). All F₁, parents, and individuals of mapping populations were measured for plant height; DJY1 and IL-87 were measured for panicle length and internode length. The yield-related traits, including tiller number, effective tiller number, number of primary branches, number of secondary branches, grain number per panicle, setting percentage and 1000-grain weight (g), were all investigated in DJY1 and NIL-2 at the mature stage. Fully filled grains were chosen randomly from each plant for measurement of 1000-grain weight.

1.3. Molecular markers and linkage map

Genomic DNA was extracted from fresh leaves of individual plants and the parents following the method of Edwards et al. (1991). Simple sequence repeat (SSR) markers were selected from the published molecular map of rice (McCouch et al., 2002). The SSR analysis was conducted in accordance with the method of Wu and Tanksley (1993). The genetic linkage map was constructed using QTL IciMapping version 3.2 software (Institute of Crop Science Chinese Academy of Agricultural Sciences, CAAS) with a minimum LOD score of 2.5.

1.4. Data analysis

SPSS version 18 in IBM was used to analyze data. The frequency distribution of plant height in the mapping populations was analyzed. The LSD multiple comparative analysis and the Student's t-test were used to analyze those data. The threshold of $\alpha = 0.05$ was considered to be statistically significant, and values of $P < 0.01$ were considered to be highly significant. The QTL controlling plant height was identified by interval mapping analysis using the QTL IciMapping software package.

2. Results

2.1. Phenotypic characteristics of IL-87

To identify phenotypic characteristics of semi-dwarf IL-87, KWH, and DJY1, plant height was measured at the maturity stage (Fig. 1A). There was a significant reduction in plant height of IL-87 (71.97 cm) compared to DJY1 (103.09 cm); specifically, IL-87 plants were 70% of the height of DJY1 plants (Fig. 1A and C). To further determine the changes in plant heights of IL-87, internode length and panicle length of IL-87 and DJY1 plants were measured at maturity. Internode length in IL-87 was significantly shorter than in DJY1 (Fig. 1B and E). In IL-87 plants the average length of each successive internode from the tip to the ground level 18.67, 10.43, 6.24, and 2.05 cm respectively. In contrast, in DJY1 plants, the successive internode lengths were 33.15, 17.65, 11.39, and 4.74 cm respectively. Average panicle length of IL-87 and DJY1 plants were 18.39 cm and 20.85 cm respectively (Fig. 1B and E). These results revealed that IL-87 has a semi-dwarf phenotype.

2.2. Genetic analysis

DJY1 was crossed with IL-87 to analyze the genetic model of the semi-dwarf phenotype. The plant height of F_1 progeny was significantly shorter than that of DJY1 but was taller than IL-87 (Fig. 1D), showing an intermediate plant height. These results indicated that the semi-dwarf trait was a semi-dominant trait. The frequency distribution of plant height of the BC_3F_2 population (2015H3E483) showed a continuous and trimodal distribution similar to that of the two parents and the intermediate types (Fig. 2A); the BC_3F_3 population (2017J1E87) also showed a similar distribution (Fig. 2B). These observations suggested that the semi-dwarf trait was controlled by a major QTL.

2.3. QTL mapping of the novel semi-dwarf trait

A total of 452 SSR markers, relatively uniformly distributed throughout the rice genomes, were used to assess polymorphism between IL-87 and DJY1. Fifteen SSR markers were polymorphic

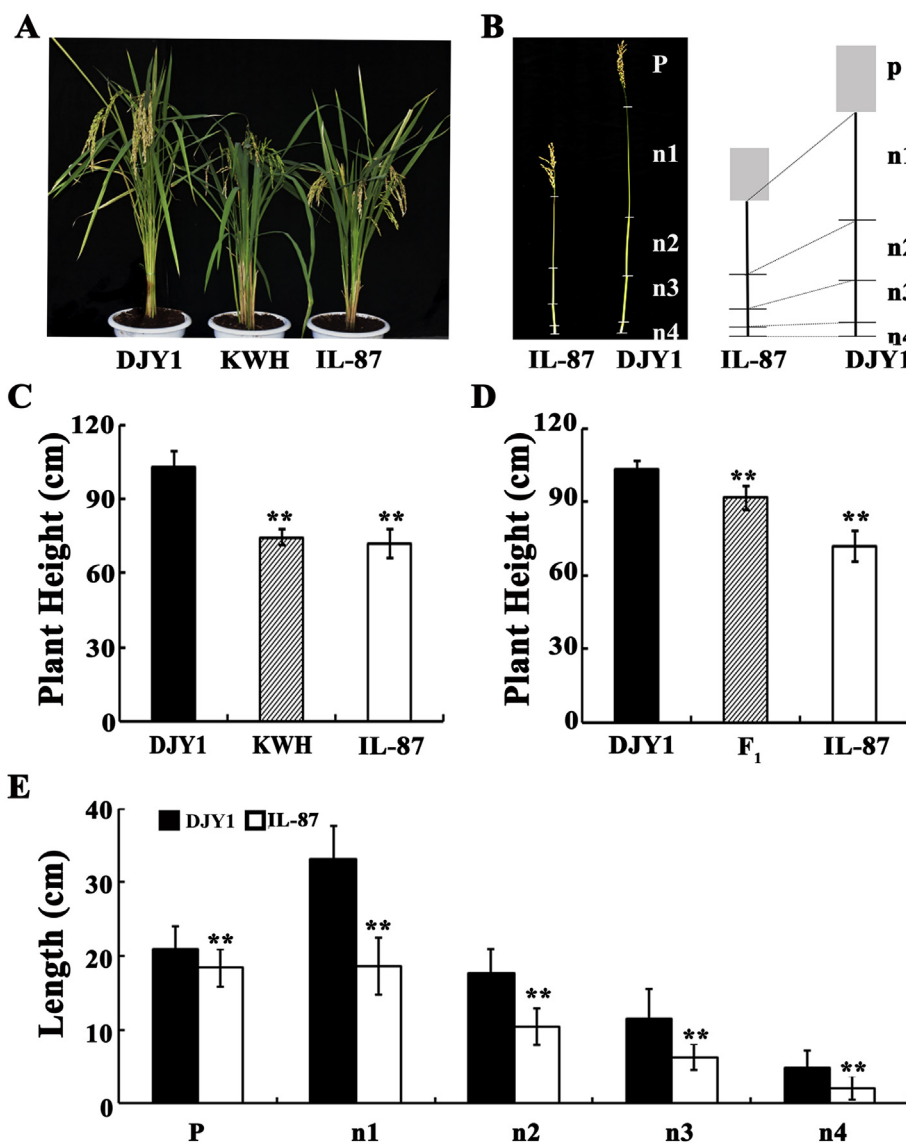


Fig. 1. Phenotypic characteristics of IL-87 and F_1 progeny. (A) Representative whole plant photos of DJY1, KWH, and IL-87 after panicle heading. (B) Comparison of the individual regions of the main stems in DJY1 and IL-87, P, panicle, n, internode. (C) Plant height of DJY1, KWH, and IL-87, as determined by measuring the main tillers. (D) Plant height of DJY1, IL-87, and F_1 , as determined by measuring the main tillers. (E) Length of the panicle and internodes, as determined using the main stems of DJY1 and IL-87, P panicle, n internode, n = 3, ** indicates extremely significant difference compared with DJY1 ($P < 0.01$); * indicates significant differences compared with DJY1 ($P < 0.05$).

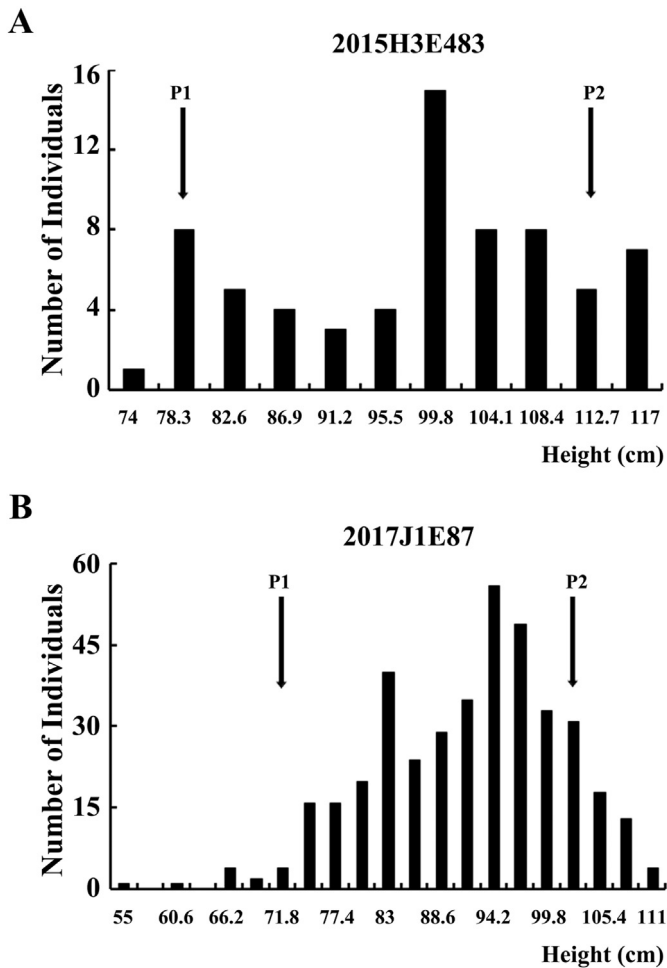


Fig. 2. Frequency distribution of plant height in the mapping populations. (A) Distribution of plant height in the BC₃F₂ population (2015H3E483). (B) Distribution of plant height in the BC₃F₃ population (2017J1E87). P1 represents the male parent (KWH), P2 represents the female parent (DJY1).

and were distributed on chromosomes 1, 4, 10, and 11 in IL-87, which was indicative of genomic introgression from KWH. The genotype of 68 individuals from the BC₃F₂ population (2015H3E483) was investigated using the 15 polymorphic SSR markers. QTL analysis showed that a QTL between RM486 and RM12057 on chromosome 1 was detected, and explained 64.39% of the phenotypic variance in plant height, and the QTL was designated *qDH1* (Fig. 3A; Table 1; Table S2).

To further localize the target QTL, a larger BC₃F₃ population (2017J1E87) containing 396 individuals derived from the original BC₃F₂ population (2015H3E483) was constructed by three heterozygous recombinant plants between RM486 and RM12057. Sixty-two SSR markers were designed in the interval between RM486 and RM12057. Among them, 5 markers were polymorphic between DJY1 and IL-87 and used for QTL analysis of 396 individuals in the BC₃F₃ population (2017J1E87). The QTL *qDH1* was further restricted to a 5.2 cM region flanked by RM6696 and RM12047 (Fig. 3B; Table S2).

To confirm that *qDH1* was the major QTL controlling the semi-dwarf phenotype, we used the BC₃F₃ mapping population to develop a NIL-2 line that was homozygous for *qDH1* at the chromosomal segment between RM6696 and RM12047. The plant heights of NIL-2 and DJY1 were assessed at maturity. The plant height of NIL-2 was significantly reduced compared with that of

DJY1, with no significant difference detected between NIL-2 and IL-87 (Fig. 3C). These results illustrate that *qDH1* is the major QTL that controlled plant height.

2.4. Yield of DJY1 and NIL-2

To investigate whether the reduction in plant height had a side effect on yield, several yield-related traits were investigated between DJY1 and NIL-2 at the maturity stage. The number of grains per panicle and secondary branches of NIL-2 (57.9 and 6.10 respectively) were significantly reduced compared with those of DJY1 (83.7 and 12.04 respectively). The average 1000-grain weight of the NIL-2 (25.61 g) was also significantly lower than that of DJY1 (30.52 g). However, the number of tillers and effective tillers were significantly increased in the NIL-2 (15.2 and 13.2, respectively) compared with those in DJY1 (9.2 and 8.6, respectively). No significant differences in the number of primary branches and setting percentage were detected. The yield of a single plant of NIL-2 (15.62 g) was less than that of DJY1 (20.13 g), with no statistically significant difference observed (Table 2). These results suggest that the negative effects of *qDH1* on rice yield are negligible.

2.5. *qDH1* affects plant height in different types of rice cultivars

To further estimate the effect of *qDH1* on rice height, NIL-2 was crossed with 20 core cultivars obtained from the IRRI, and 18 F₁ progenies were raised (two crosses failed). All F₁ progeny were planted under the same conditions in our experimental station and phenotyped for plant height. Among the 18 crosses, the plant height of F₁ progeny in nine crosses decreased, whereas the plant height of the F₁ progeny in the remaining nine cross groups was unchanged. No significant difference in plant height was observed between the F₁ progenies and the parent 'IR64-21', 'M 202', 'Minghui 63' (MH63), 'N 22', 'Nipponbare' (Npb), 'Sadu-Cho' (Sadu), 'Shan-Huang-Zhan-2' (SH22), 'Zhenshan 97B' (ZS97B) and 'Cypress' (Cyp) (Fig. 4A and C). Plant height of F₁ progeny was highly significantly reduced compared with that of the parent 'Azucena' (Azu), 'Dom-Sufid' (Doms), 'FR13 A', 'Li-Jiang-Xin-Tuan-Hei-Gu' (LTH), and 'Pokkali's' (Pok), by about 30%, 20%, 18%, 33%, and 37%, respectively (Fig. 4A and B; Table S3). The F₁ progenies of 'Dular', 'Moroberekan' (Moro), 'Tainung 67' (TNG67), and 'Aswina' (Asw) were significantly shorter than the parent by 27%, 10%, 11%, and 21%, respectively (Fig. 4A and B; Table S3). Among them, Azu, Moro, LTH, TNG67 and Doms belonged to the *Japonica* group, 'FR13 A' and 'Dular' belonged to the *Aus* group, Asw and Pok were attributed to *Indica* group (Table S1). These results indicate that homozygous NIL-2 can potentially reduce plant height in different types of rice cultivars from different genetic background.

3. Discussion

Plant height is an important target of crop breeding and is positively correlated with yield within a certain range. A suitable plant height is not only beneficial for the ideal plant type, but also plays an important role in improving the yield potential of rice and the photosynthetic efficiency of the upper leaves (Luo et al., 2012). Since the use of the *sd1* gene in the 'Green Revolution' of rice, an increasing number of plant height genes have been identified and cloned. However, many of them cannot be used for breeding due to unwanted phenotypes such as extreme dwarfism, sterility, and short grain (Sakamoto et al., 2004). Dominant dwarf genes are of great value and widely used in crop breeding (Liang et al., 2011). The 'Green Revolution' genes of wheat, *Rht1* and *Rht2*, are dominant dwarf genes, have been successfully used in wheat breeding (Kurkiewicz et al., 2007; Gale et al., 1985). In this study, we used the

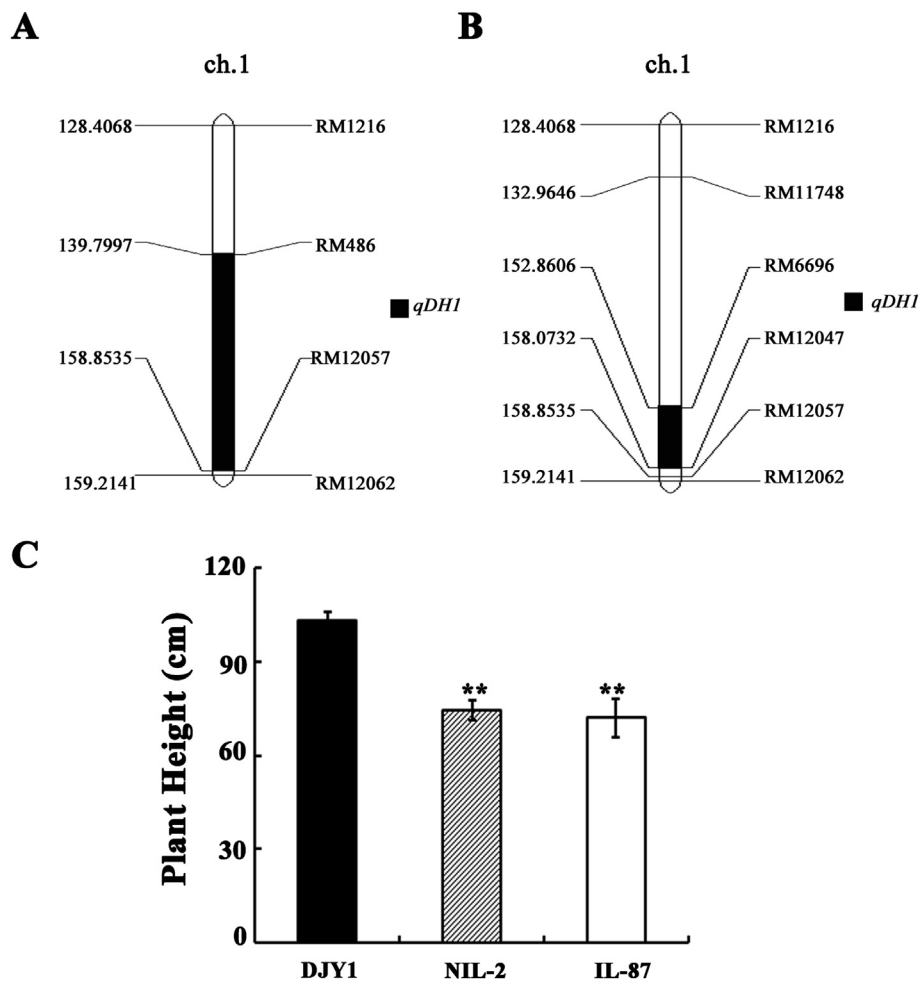


Fig. 3. Mapping of QTL for the semi-dwarf trait. (A) The results of QTL mapping based on the BC₃F₂ population (2015H3E483). (B) The results of QTL mapping based on the BC₃F₃ population (2017J1E87). (C) Plant height of DJY1, NIL-2, and IL-87, as determined by measuring the main tillers. n = 3, ** indicates extremely significant difference compared with DJY1 ($P < 0.01$).

Table 1
QTLs detected for plant height in the population after two generations.

Populations name	QTL	Flanking marker	Trait	LOD	Variance explained (%)	Additive effect
2015H3E483	<i>qDH1</i>	RM486-RM12057	height	11.9117	64.3906	10.4982
2017J1E87	<i>qDH1</i>	RM6696-RM12049	height	34.1495	35.7837	6.3784

japonica cultivar DJY1 and a KWH donor to develop the near-isogenic line NIL-2, which has the semi-dominant dwarf QTL *qDH1*. The semi-dominant semi-dwarf QTL *qDH1* was localized between the markers RM6696 and RM12047 on chromosome 1 and

we found that *qDH1* reduces rice plant height with few side effects on yield.

In recent decades, a large number of plant height QTLs have been detected in the rice genome. Bao et al. (2009) used

Table 2
Yield of DJY1 and NIL-2.

Traits	DJY1	NIL-2	P-value
Number of grain per panicle	83.70 ± 19.06	57.90 ± 9.11*	0.0211
Setting percentage	79.44% ± 8.91%	81.01% ± 8.29%	0.7788
Number of tillers	9.20 ± 1.48	15.20 ± 4.66*	0.0252
Number of effective tillers	8.60 ± 1.67	13.20 ± 2.58*	0.0102
1000-grain weight (g)	30.52 ± 1.72	25.61 ± 0.46**	0.0001
Number of primary branches	8.57 ± 1.22	7.43 ± 0.54	0.0858
Number of secondary branches	12.04 ± 4.02	6.10 ± 1.97*	0.0141
Theoretical yield of single plant (g)	20.13 ± 3.49	15.62 ± 3.33	0.0701

N = 3, ** $P < 0.01$ by Student's t-test; * $P < 0.05$ by Student's t-test.

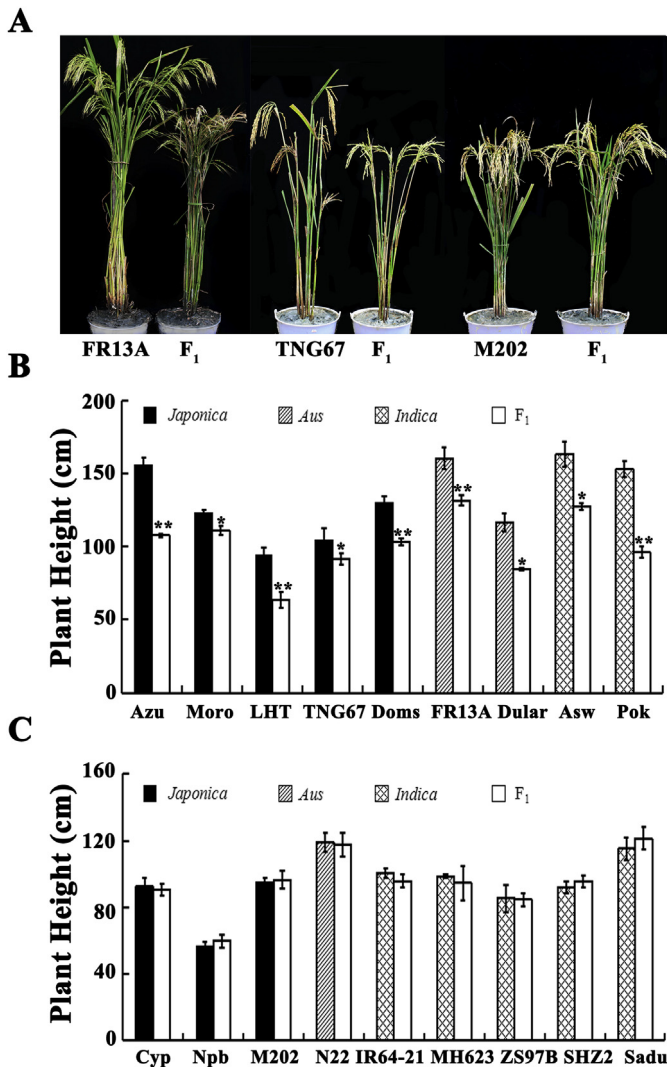


Fig. 4. Evaluation of breeding value. (A) Representative plant height phenotype. (B) (C) Height analysis of each parent and F₁ generation. n = 3, ** indicates extremely significant differences between F₁ and Parents ($P < 0.01$); * indicates significant differences between F₁ and parent ($P < 0.05$).

populations derived from residual heterozygous lines to detect QTLs for plant height in rice, and localized the QTL *qph6-1* on chromosome 6 to a physical distance of about 51.7 KB. The QTL *ph1* was identified on chromosome 1 of rice and cloned, the gene was reported highly correlated with plant height (Ishimaru et al., 2004). *HEADING DATE 7* (*GHD7*), which controls the number of single panicles, plant height, and the timing of heading of rice, has been cloned. The gene encodes a CCT domain-containing protein and is mainly regulated by photoperiod (Xue et al., 2008). In the present study, we used different generations of the same genetic population to narrow down the QTL locus *qDH1* for plant height between the markers RM6696 and RM12047 on chromosome 1. These results provide a sound foundation for further fine mapping, cloning, and functional analysis of the locus.

The use of dwarf genes is important for the breeding of hybrid rice. Although breeders are constantly seeking to create novel dwarf germplasm by a variety of methods, few dwarf genes have been exploited for rice breeding. For example, the dwarf germplasm of japonica, which is mainly derived from 'Nonken 58' and 'Bali11a', is rarely used for breeding because of poor economic characteristics (Chen et al., 2013). We found that the major QTL

qDH1 induces reduction of plant height in DJY1, but has no negative effects on yield (Fig. 1B; Table 2). Hybridization of NIL-2 with 20 core rice cultivars belonging to three genetic background groups (Table S1), and assessment of the plant height of the F₁ progeny showed that *qDH1* could reduce the plant height of nine cultivars from different genetic background. Because plant height is a phenotype controlled by multiple genes (Ni et al., 2009; Jaiswal et al., 2002), the interactions between these genes determine plant height differences (Wade, 2001). This may explain why plant height was affected in only half the cultivars examined.

The well-characterized semi-dwarf gene *sd1* was also located between the markers RM6696 and RM12047 on chromosome 1. *sd1* is a recessive gene located on the long arm of chromosome 1 and reduces rice plant height with few side effects on yield (Hedden, 2003). The QTL *qDH1* also presents a similar phenotype. Thus, we speculate that *qDH1* might be the dominant allele of *sd1*. However, in our study, the QTL *qDH1* was semi-dominant. If QTL *qDH1* is an allele of *sd1*, it will be important for studying the molecular mechanisms by which different alleles change from recessive to dominant.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.pld.2018.09.001>.

References

- Asano, K., Hirano, K., Ueguchi-Tanaka, M., et al., 2009. Isolation and characterization of dominant dwarf mutants, *Slr1-d*, in rice. *Mol. Genet. Genom.* 281, 223–231.
- Ayano, M., Kani, T., Kojima, M., et al., 2014. Gibberellin biosynthesis and signal transduction is essential for internode elongation in deepwater rice. *Plant Cell Environ.* 37, 2313–2324.
- Bao, Q.J., Fan, Y.Y., Yu, W.D., 2009. Fine mapping of QTL *qPH6-1* for plant height on the short arm of rice chromosome 6. *Chin. J. Rice Sci.* 23, 470–474.
- Chen, F., Lin, H.Z., Zhou, Q.X., et al., 2013. Research progress of dwarf genes in rice. *Shandong Agric. Sci.* 45, 127–133.
- Edwards, K., Johnstone, C., Thompson, C., 1991. A simple and rapid method for the preparation of plant genomic DNA for PCR analysis. *Nucleic Acids Res.* 19, 1349.
- Gale, M.D., Youssefian, S., Russell, G.E., 1985. *Dwarfing Genes in Wheat*. Elsevier Inc, London.
- Hargrove, T.R., Cabanilla, V.L., 1979. The impact of semidwarf varieties on Asian rice-breeding programs. *Bioscience* 29, 731–735.
- Hedden, P., 2003. The genes of the green revolution. *Trends Genet.* 09, 5–9.
- Hong, Z., Ueguchi-Tanaka, M., Shimizu-Sato, S., et al., 2002. Loss-of-function of a rice brassinosteroid biosynthetic enzyme, C-6 oxidase, prevents the organized arrangement and polar elongation of cells in the leaves and stem. *Plant J.* 32, 495–508.
- Ishimaru, K., Ono, K., Kashiwagi, T., 2004. Identification of a new gene controlling plant height in rice using the candidate-gene strategy. *Planta* 218, 388–395.
- Itoh, H., Ueguchi-Tanaka, M., Sentoku, N., 2001. Cloning and functional analysis of two gibberellin 3 β -hydroxylase genes that are differently expressed during the growth of rice. *P.N.A.S.* 98, 09–14.
- Iwamoto, M., Kiyota, S., Hanada, A., et al., 2011. The multiple contributions of phytochromes to the control of internode elongation in rice. *Plant Physiol.* 157, 1187–1195.
- Jaiswal, P., Ware, D., Ni, J., et al., 2002. Gramene: development and integration of trait and gene ontologies for rice. *Comp. Funct. Genom.* 3, 132–136.
- Kagiyama, M., Hirano, Y., Mori, T., et al., 2013. Structures of D14 and D14L in the strigolactone and karrikin signaling pathways. *Gene Cell.* 18, 147–160.
- Khush, G.S., 1999. Green revolution: preparing for the 21st century. *Genome* 42, 646–655.
- Khush, G.S., 2005. What it will take to feed 5.0 billion rice consumers in 2030. *Plant Mol. Biol.* 59, 1–6.
- Kurkiewicz, K.U., 2007. Introgression of the reduced height gene *Rht10* from the wheat cultivar Ai-bian 1 into the triticale genotype. *Genetika* 43, 1269–1272.

- Liang, F., Xin, X.Y., Hu, Z.J., et al., 2011. Genetic analysis and fine mapping of a novel semidominant dwarfing gene LB4D in rice. *J. Integr. Plant Biol.* 53, 312–323.
- Liang, G.H., Cao, X.Y., Sui, J.M., et al., 2004. Fine mapping of a semidwarf gene sd-g in indica rice (*Oryza sativa* L.). *Chin. Sci. Bull.* 49, 900–904.
- Liu, B.M., Wu, Y.J., Fu, X.D., et al., 2008. Characterizations and molecular mapping of a novel dominant semi-dwarf gene Sdd (t) in rice (*Oryza sativa*). *Plant Breed.* 127, 125–130.
- Liu, C., Zheng, S., Gui, J.S., et al., 2018. Shortened basal internodes encodes a gibberellin 2-oxidase and contributes to lodging resistance in rice. *Mol. Plant* 11, 288–299.
- Liu, W.Z., Wu, C., Fu, Y.P., et al., 2009. Identification and characterization of HTD2: a novel gene negatively regulating tiller bud outgrowth in rice. *Planta* 230, 649–658.
- Lu, Z.F., Yu, H., Xiong, G.S., et al., 2013. Genome-wide binding analysis of the transcription activator ideal plant architecture1 reveals a complex network regulating rice plant architecture. *Plant Cell* 25, 3743–3759.
- Luh, B.S., 1980. *Rice: Production and Utilization*. AVI Publishing Co., Inc, Westport, CT.
- Luo, J., Shao, G.N., Wei, X.J., et al., 2012. Genetic analysis of a QTL qPH3 for plant height in rice. *Chin. J. Rice Sci.* 26, 417–442.
- Matsuo, T., Futsuhara, Y., Kikuchi, F., et al., 1997. *Science of the Rice Plant. 3. Genetics*. Supplementary Volume: Indices. Food and Agriculture Policy Research Center, Tokyo.
- McCouch, S.R., Teytelman, L., Xu, Y., et al., 2002. Development and mapping of 2240 new SSR markers for rice (*Oryza sativa* L.) (supplement). *DNA Res.* 9, 199–207.
- McNally, K.L., Childs, K.L., Bohnert, R., et al., 2009. Genomewide SNP variation reveals relationships among landraces and modern varieties of rice. *P.N.A.S. USA* 106, 12273–12278.
- Miura, K., Wu, J., Sunohara, H., et al., 2009. High-resolution mapping revealed a 1.3-Mbp genomic inversion in Ssi1, a dominant semidwarf gene in rice (*Oryza sativa*). *Plant Breed.* 128, 63–69.
- Miura, K., Ikeda, M., Matsubara, A., et al., 2010. OsSPL14 promotes panicle branching and higher grain productivity in rice. *Nat. Genet.* 42, 545–549.
- Mori, M., Nomura, T., Ooka, H., et al., 2002. Isolation and characterization of a rice dwarf mutant with a defect in brassinosteroid biosynthesis. *Plant Physiol.* 130, 1152–1161.
- Nakamura, A., Fujioka, S., Sunohara, H., et al., 2006. The role of OsBRI1 and its homologous genes, OsBRL1 and OsBRL3, in rice. *Plant Physiol.* 140, 580–590.
- Ni, J.J., Pujar, A., Youens-Clark, K., et al., 2009. Gramene QTL database: development, content and applications. *Database J. Biol. Databases*. Curation 1, bap005.
- Piao, R.H., Chu, S.H., Jiang, W.Z., et al., 2014. Isolation and characterization of a dominant dwarf gene, D-h, in rice. *PLoS One* 9, e86210.
- Qin, R.Z., Qiu, Y., Cheng, Z.J., et al., 2008. Genetic analysis of a novel dominant rice dwarf mutant 986083D. *Euphytica* 160, 379–387.
- Sakamoto, T., Miura, K., Itoh, H., et al., 2004. An overview of gibberellin metabolism enzyme genes and their related mutants in rice. *Plant Physiol.* 135, 1642–1653.
- Sunohara, H., Kitano, H., 2003. New dwarf mutant controlled by a dominant gene, Twisted dwarf1. *Rice Genet. Newsl.* 20, 20–21.
- Tanaka, N., Matsuoka, M., Kitano, H., et al., 2006. Gid1, a gibberellin-insensitive dwarf mutant, shows altered regulation of probenazole-inducible protein (PBZ1) in response to cold stress and pathogen attack. *Plant. Plant Cell Environ.* 29, 619–631.
- Tong, H.N., Xiao, Y.H., Liu, D.P., et al., 2014. Brassinosteroid regulates cell elongation by modulating gibberellin metabolism in rice. *Plant Cell* 26, 4376–4393.
- Tong, J.P., Liu, X.J., Zhang, S.Y., et al., 2007. Identification, genetic characterization, GA response and molecular mapping of Sdt97: a dominant mutant gene conferring semi-dwarfism in rice (*Oryza sativa* L.). *Genet. Res.* 89, 221–230.
- Ueguchi-Tanaka, M., Ashikari, M., Nakajima, M., et al., 2005. GIBBERELLIN INSENSITIVE DWARF1 encodes a soluble receptor for gibberellin. *Nature* 437, 693–698.
- Wade, M.J., 2001. Epistasis, complex traits, and mapping genes. *Genetica* 112–113, 59–69.
- Wang, J.G., Gang, S., Yang, L., et al., 2017. Markers associated with culm length and elongated internode length in japonica, rice. *Crop Sci.* 57, 2329–2344.
- Wei, L.R., Xu, J.C., Li, X.B., et al., 2006. Genetic analysis and mapping of the dominant dwarfing gene D-53 in rice. *J. Integr. Plant Biol.* 48, 447–452.
- Wu, K.S., Tanksley, S.D., 1993. Abundance, polymorphism and genetic mapping of microsatellites in rice. *Mol. Gen. Genet.* 241, 225–235.
- Würschum, T., Langer, S.M., Longin, C.F.H., 2015. Genetic control of plant height in European winter wheat cultivars. *Theor. Appl. Genet.* 128, 865.
- Xue, W.Y., Xing, Y.Z., Weng, X.Y., et al., 2008. Natural variation in Ghd7 is an important regulator of heading date and yield potential in rice. *Nat. Genet.* 40, 761–767.
- Yamamoto, C., Ihara, Y., Wu, X., et al., 2000. Loss of function of a rice brassinosteroid insensitive1 homolog prevents internode elongation and bending of the lamina joint. *Plant Cell* 12, 1591–1606.
- Zhang, B.W., Wang, X.L., Zhao, Z.Y., et al., 2016. OsBRI1 activates BR signaling by preventing binding between the TPR and kinase domains of OsBSK3 via phosphorylation. *Plant Physiol.* 170, 1149–1161.
- Zhang, J., Liu, X.Q., Li, S.Y., et al., 2014. The rice semi-dwarf mutant sd37, caused by a mutation in CYP96B4, plays an important role in the fine-tuning of plant growth. *PLoS One* 9, e88068.
- Zhang, L., Yu, H., Ma, B., et al., 2017a. A natural tandem array alleviates epigenetic repression of IPA1 and leads to superior yielding rice. *Nat. Commun.* 8, 14789.
- Zhang, Y.X., Yu, C.S., Lin, J.Z., et al., 2017b. OsMPH1 regulates plant height and improves grain yield in rice. *PLoS One* 12, e0180825.
- Zhao, C.F., Zhao, Q.Y., Zhao, L., et al., 2018a. Characterization and fine mapping of qPE12, a new locus controlling rice panicle exertion. *Euphytica* 214, 47.
- Zhao, Z.G., Zhang, C., Liu, X., et al., 2018b. Genetic analysis and fine mapping of a dominant dwarfness gene from wild rice (*Oryza barthii*). *Plant Breed.* 137, 50–59.